

The Economic KKT Cookbook

A portable method for constrained optimization, corners, shadow prices, and comparative statics

What this booklet gives you

One sign convention. One active-set algorithm. One verification checklist. The method applies to consumer choice, firms, portfolios, mechanism constraints, and dynamic problems solved one state at a time. A separate algebraic toolkit shows when structure can solve most of the problem before the multipliers appear.

Notation	Write every inequality as a nonnegative slack before assigning a multiplier.
Corners	Replace unjustified equalities by complementarity and sign checks.
Shortcuts	Use tangent directions, bounds, homogeneity, duality, and projection before brute force.
Interpretation	Read multipliers as marginal values only after the candidate is verified.
Examples	Expected-utility portfolio choice and a Galor-inspired quantity-quality model.

Victor H. Aguiar

Simon Fraser University | Fall 2026

Student edition. The examples are fully worked so that the booklet can be used for review, office hours, and core-exam preparation.

START HERE The method in sixty seconds

The Karush-Kuhn-Tucker (KKT) conditions extend the Lagrange method to inequality constraints and economically important corner solutions.

Consider a maximization problem with nonnegative choices and constraints written as nonnegative slacks:

$$\max_{x \geq 0} f(x; \theta) \text{ subject to } s_j(x; \theta) \geq 0, h_k(x; \theta) = 0.$$

Use multipliers $\lambda_j \geq 0$ on $s_j \geq 0$ and unrestricted multipliers ν_k on equalities. Define the **net marginal payoff**

$$D_i(x, \lambda, \nu) = f_i(x) + \sum_j \lambda_j s_{j,i}(x) + \sum_k \nu_k h_{k,i}(x).$$

1. Normalize. Rewrite every inequality as a slack $s_j(x) \geq 0$.

2. Write complementarity.

$$0 \leq x_i \perp -D_i \geq 0, 0 \leq \lambda_j \perp s_j(x) \geq 0.$$

3. Guess an active set. Which choices are positive? Which constraints bind?

4. Solve equalities. Positive x_i gives $D_i = 0$; binding s_j gives $s_j = 0$.

5. Audit every sign. Check feasibility, multiplier signs, omitted-choice inequalities, slack constraints, and every complementary product.

6. Certify and interpret. Use concavity for global optimality; then use multipliers and the envelope theorem for economics.

At a positive choice, marginal benefit equals its shadow opportunity cost. At a zero choice, marginal benefit may be lower. This is why an equality-only FOC method misses economically important corners.

How to use this booklet

Pages 3–8 build the method. Pages 9–16 develop an algebra-before-KKT toolkit. Pages 17–19 solve portfolio choice, and pages 20–23 solve the quantity-quality example. Page 24 is a mastery protocol; page 25 is a tear-out reference sheet.

Never memorize a multiplier sign in isolation. First choose a slack direction. This booklet always uses $s(x) \geq 0$ and adds $+\lambda s(x)$ to a maximization Lagrangian. Changing convention is harmless only if every associated sign changes with it.

1 | TRANSLATE Put the economics into one canonical form

1 Before KKT: model the feasible set

KKT is easiest after the economic constraints are translated into one language. For a maximization problem, this booklet uses **nonnegative slacks**:

$$s_j(x) \geq 0.$$

The multiplier $\lambda_j \geq 0$ is attached to $+\lambda_j s_j(x)$.

Economic statement	Nonnegative slack	Interpretation of multiplier
$p \cdot x \leq m$	$m - p \cdot x \geq 0$	marginal value of one more unit of income
$q \leq F(z)$	$F(z) - q \geq 0$	marginal value of relaxing technology
$e_i \geq \underline{e}_i$	$e_i - \underline{e}_i \geq 0$	marginal value of lowering the requirement
$Ax \leq b$	$b - Ax \geq 0$	marginal value of the relevant right-hand-side resource
$x_i \geq 0$	keep as a sign restriction	forgone gain from forcing the choice to remain at zero
$h(x) = 0$	equality, no slack	multiplier is unrestricted in sign

For vectors a, b of the same length,

$$0 \leq a \perp b \geq 0$$

means $a \geq 0, b \geq 0$, and $a_i b_i = 0$ for every component i . Thus at least one member of each pair is zero. Both may be zero.

Units are a fast error check

If f is measured in utility and a slack is measured in dollars, its multiplier has units “utility per dollar.” Rescaling a constraint rescales its multiplier. For example, $m - px \geq 0$ and $2m - 2px \geq 0$ describe the same feasible set, but the second multiplier is half the first. Economic claims about shadow prices must therefore report the normalization.

Before writing a Lagrangian, identify the choice variables, domains, economic constraints, and parameter units. If a bound is already implied by other constraints, do not duplicate it: redundant constraints create redundant and often nonunique multipliers.

2 | DERIVE Full KKT and the compact economic form

2 One derivation worth knowing

Take

$$\max_{x \geq 0} f(x) \text{ subject to } s(x) \geq 0, h(x) = 0.$$

Introduce $\lambda \geq 0$ for $s(x) \geq 0$, $\mu \geq 0$ for $x \geq 0$, and a free vector ν for $h(x) = 0$:

$$\mathcal{L}(x, \lambda, \nu, \mu) = f(x) + \lambda^\top s(x) + \nu^\top h(x) + \mu^\top x.$$

The full stationarity condition is

$$\nabla f(x) + J_s(x)^\top \lambda + J_h(x)^\top \nu + \mu = 0.$$

Let

$$D := \nabla f(x) + J_s(x)^\top \lambda + J_h(x)^\top \nu.$$

Then stationarity says $\mu = -D$. Since $\mu \geq 0$ and $x_i \mu_i = 0$, we can eliminate μ and write the compact system

$$0 \leq x \perp -D \geq 0, 0 \leq \lambda \perp s(x) \geq 0, h(x) = 0. \quad (1)$$

For each choice x_i :

$$x_i > 0 \implies D_i = 0,$$

$$x_i = 0 \implies D_i \leq 0.$$

The inequality says that a choice held at zero cannot have a positive net marginal payoff. Equality is required only for a positive choice.

What if a variable is free in sign?

If x_i may be positive or negative, do not attach a nonnegativity multiplier. Use the equality $D_i = 0$. If a sign is already implied by other constraints, you may also treat the variable as sign-free and avoid a redundant multiplier, provided the implication is exact.

The component statement $\nabla f - A^\top \lambda \leq 0$ applies to choices restricted by $x \geq 0$. It is not the stationarity condition for an unrestricted choice, which requires equality.

3 | REMEMBER The linear-resource formula

3 The formula used most often in economics

For

$$\max_{x \geq 0} f(x) \text{ subject to } Ax \leq b,$$

use the slack $b - Ax \geq 0$ and multiplier $\lambda \geq 0$. Equation (1) becomes

$$\begin{aligned} b - Ax &\geq 0, \lambda \geq 0, \lambda_j(b_j - A_jx) = 0, \\ \nabla f(x) - A^\top \lambda &\leq 0, \\ x_i [\nabla_i f(x) - (A^\top \lambda)_i] &= 0. \end{aligned}$$

Equivalently,

$$0 \leq x \perp A^\top \lambda - \nabla f(x) \geq 0, \quad 0 \leq \lambda \perp b - Ax \geq 0.$$

Three economic readings

- Used activity:** if $x_i > 0$, marginal benefit equals the total shadow cost $A_i^\top \lambda$.
- Unused activity:** if $x_i = 0$, marginal benefit is no larger than its total shadow cost.
- Scarce resource:** if $\lambda_j > 0$, resource j is exhausted. A resource can be exhausted with $\lambda_j = 0$.

Complementarity gives

$$\lambda_j > 0 \implies b_j - A_jx = 0.$$

It does not give the reverse implication. A binding constraint can be locally valueless, and therefore have a zero multiplier. The second worked example shows this at the apex of a quality cone.

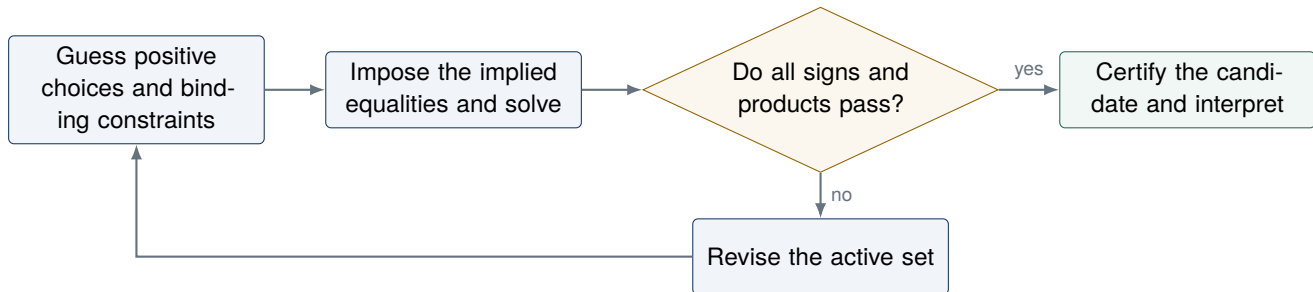
Equality-constrained resources

If a resource constraint is written as $Ax = b$, its multiplier is free in sign and stationarity is an equality for unrestricted choices. In many monotone consumer or firm problems it is cleaner first to prove that $Ax \leq b$ binds, then substitute. Preserve the reduced domain and record how reduced multipliers map back to the original problem.

4 | SOLVE The active-set algorithm

4 Turn complementarity into ordinary algebra

KKT looks nonlinear because each complementary pair has two possible branches. Economic reasoning often identifies the likely branch. An **active set** is simply a guess about which choices are positive and which constraints bind.



Choose sets P and Z of positive and zero choices, and sets B and S of binding and slack constraints. Solve

$$D_i = 0 \quad (i \in P), \quad x_i = 0 \quad (i \in Z),$$

$$s_j(x) = 0 \quad (j \in B), \quad \lambda_j = 0 \quad (j \in S).$$

Then verify

$$x_P > 0, \quad -D_Z \geq 0, \quad \lambda_B \geq 0, \quad s_S(x) > 0.$$

Weak inequalities are allowed when the solution is degenerate.

The final audit

- ✓ Every original constraint is feasible.
- ✓ Every inequality multiplier has the declared sign.
- ✓ Every positive choice has zero residual; every zero choice has a nonnegative residual.
- ✓ Every slack constraint has zero multiplier.
- ✓ Every complementary product is zero.
- ✓ A theorem, not optimism, justifies the global claim.

5 | CERTIFY

What KKT does and does not prove

5 Separate four logically different questions

Suppose f , the slacks, and the equality constraints are differentiable. Let f and every slack s_j be concave, and let every h_k be affine. Then every feasible point that satisfies the KKT system is a global maximizer.

The reason is geometric: the first-order concavity bound on f and on each slack, combined with multiplier nonnegativity and complementarity, rules out a feasible direction with higher objective value.

Question	What is needed?
Does an optimizer exist?	Compact feasible set plus continuity, or another coercivity/existence argument. KKT alone does not give existence.
Must an optimizer satisfy KKT?	A constraint qualification. Slater’s condition is the standard convenient condition for convex feasible problems.
Is a KKT point globally optimal?	Concave objective, concave nonnegative slacks, and affine equalities. Slater is not needed for this direction.
Is the optimizer unique?	Strict concavity of the objective on the feasible set. This does not imply unique multipliers.

For the present convention, a Slater point satisfies every nonlinear inequality strictly, $s_j(x) > 0$, while satisfying all equalities. Under the usual convexity and regularity assumptions, it guarantees strong duality and the existence of KKT multipliers at an optimum. It is a necessity tool, not an existence theorem for the primal optimizer.

If the objective or constraints are nonsmooth, replace gradients by subgradients or normal cones. If the problem is nonconcave, KKT points are candidates: compare their objective values with boundaries and other stationary points. If an optimum has zero consumption under $u(c) = \sqrt{c}$ or $\log c$, ordinary derivatives may fail; use domain arguments or nonsmooth methods.

6 | INTERPRET

Envelope-theorem magic, used carefully

6 Comparative statics without differentiating the policy

Let

$$V(\theta) = \max_x f(x; \theta) \text{ subject to } s(x; \theta) \geq 0, \ h(x; \theta) = 0.$$

At a regular point where the value function is differentiable, the envelope formula is

$$V_\theta = f_\theta(x^*; \theta) + \lambda^{*\top} s_\theta(x^*; \theta) + \nu^{*\top} h_\theta(x^*; \theta).$$

(2)

There is no $dx^*/d\theta$ term. Optimality makes the first-order effect of the induced policy change vanish.

Problem	Envelope result	Economic reading
$Ax \leq b$	$V_{b_j} = \lambda_j^*$	value of one more unit of resource j
Consumer budget $p \cdot x \leq m$	$v_m = \lambda^*$	marginal utility of wealth
Same consumer problem	$v_{p_i} = -\lambda^* x_i^*$	price effect holding the policy fixed
Profit with capacity $q \leq \bar{q}$	$\pi_{\bar{q}} = \lambda^*$	value of one more unit of capacity

If $\lambda^* > 0$, the consumer formulas imply Roy’s identity:

$$x_i^*(p, m) = -\frac{v_{p_i}(p, m)}{v_m(p, m)}.$$

First solve and verify the optimization problem. Then use the multiplier to value a small relaxation. The envelope theorem avoids solving explicitly for how every choice changes with the parameter. It is an interpreter, not a solver.

At an active-set switch, the value function may have a kink. A single derivative may not exist; use one-sided or directional derivatives, or the relevant range of multipliers. Also remember that multiplying a constraint by a positive constant inversely rescales its multiplier.

7 | ECONOMIZE

Discover first; certify second

7 KKT is a certificate, not an equation factory

An expert normally solves the same problem along two paths. The **discovery path** uses economics, algebra, geometry, or an inequality to locate a candidate. The **certification path** uses KKT to prove that the candidate leaves no profitable feasible direction and to recover shadow prices.

For linear resources define the primal slack and the reduced-cost residual

$$s = b - Ax, \qquad r = A^\top \lambda - \nabla f(x).$$

Then the entire inequality logic is

$$0 \leq \lambda \perp s \geq 0, \qquad 0 \leq x \perp r \geq 0.$$

Primal object	Dual mate	Economic state
$s_j > 0$	$\lambda_j = 0$	resource is abundant
$s_j = 0$	$\lambda_j > 0$	resource is scarce and locally valuable
$s_j = 0$	$\lambda_j = 0$	binding but locally valueless; often a regime boundary
$x_i > 0$	$r_i = 0$	activity is used; marginal benefit equals shadow cost
$x_i = 0$	$r_i > 0$	activity is strictly not worth entering
$x_i = 0$	$r_i = 0$	indifference, degeneracy, or a regime boundary

1. **Screen economically.** Use monotonicity, dominance, Inada conditions, and units to predict binding resources and impossible corners.

2. **Reduce dimension.** Substitute a proved binding equality, choose natural coordinates, or move along tangent directions.

3. **Exploit structure.** Look for separability, homogeneity, completing the square, ratios, or a sharp inequality.

4. **Certify.** Restore every omitted domain restriction and audit the full KKT system. Then invoke the appropriate global theorem.

A substitution is equivalent to the original problem only when its reduced domain carries every implied inequality. A clever candidate without those checks is not yet a solution.

8 | REDUCE Tangent directions and better coordinates

8 Eliminate multipliers without losing constraints

Suppose the active equalities are $Bx = d$. A feasible first-order movement z along that face satisfies $Bz = 0$. At an interior point of the face,

$$z^\top \nabla f(x^*) = 0 \quad \text{for every } z \in \ker B.$$

If the columns of N span $\ker B$, solve $N^\top \nabla f(x) = 0$ together with $Bx = d$. This is the matrix version of substituting the budget and often removes all equality multipliers.

With one budget $p \cdot x = m$, a budget-neutral movement between positive goods i and k gives

$$p_k f_i - p_i f_k = 0 \quad \Longleftrightarrow \quad \frac{f_i}{p_i} = \frac{f_k}{p_k}.$$

The familiar marginal-utility-per-dollar rule is therefore a tangent-direction condition. It applies only after positivity has been established.

Change from assets to state goods

For the portfolio example, let $c = (c_L, c_H)^\top$ and

$$c_L = \frac{1}{2}\alpha + \beta, \quad c_H = \frac{7}{4}\alpha + \beta.$$

Inverting this payoff map gives

$$\alpha = \frac{4}{5}(c_H - c_L), \quad \beta = \frac{1}{5}(7c_L - 2c_H).$$

Consequently, the asset budget becomes a state-price budget:

$$\frac{3}{5}c_L + \frac{2}{5}c_H \leq w. \tag{3}$$

No short selling survives as the consumption cone

$$c_H \geq c_L, \quad 2c_H \leq 7c_L.$$

If state consumption is $c = Rx$, asset prices are p , and R is invertible, then $x = R^{-1}c$ and the state-price vector is

$$q = R^{-\top} p.$$

The budget becomes $q \cdot c \leq w$, while portfolio restrictions become constraints on $R^{-1}c$. In incomplete markets, R is not invertible and the attainable-consumption subspace must remain explicit.

9 | BOUND A global solution without differentiation

9 Cauchy–Schwarz solves square-root utility

Consider state prices $q_s > 0$, probabilities $\pi_s > 0$, and $\sum_s q_s c_s \leq w$. Then

$$\sum_s \pi_s \sqrt{c_s} = \sum_s \frac{\pi_s}{\sqrt{q_s}} \sqrt{q_s c_s} \leq \sqrt{\sum_s \frac{\pi_s^2}{q_s}} \sqrt{\sum_s q_s c_s}.$$

For the portfolio state prices in (3),

$$\frac{(1/2)^2}{3/5} + \frac{(1/2)^2}{2/5} = \frac{25}{24}.$$

Thus, at $w = 24$,

$$F(c_L, c_H) \leq \sqrt{\frac{25}{24}} 24 = 5.$$

Equality in Cauchy–Schwarz requires

$$\sqrt{c_s} \propto \frac{\pi_s}{q_s}, \quad \sqrt{c_L} : \sqrt{c_H} = 2 : 3.$$

The state-price budget then gives

$$(c_L, c_H) = (16, 36), \quad (\alpha, \beta) = (16, 8).$$

The holding-cone inequalities hold, so this sharp upper bound is attainable. It proves global optimality before KKT is written.

For $0 < \rho < 1$, define

$$S = \sum_s \pi_s^{1/(1-\rho)} q_s^{-\rho/(1-\rho)}.$$

Then

$$\max_{c \geq 0, q \cdot c \leq w} \sum_s \pi_s c_s^\rho = w^\rho S^{1-\rho},$$

and

$$c_s^* = \frac{w}{S} \left(\frac{\pi_s}{q_s} \right)^{1/(1-\rho)}.$$

The square-root result is the case $\rho = 1/2$.

A sharp bound solves the relaxed state-good problem. Portfolio cones, borrowing limits, or incomplete markets still require feasibility checks. If the equality case violates them, return to KKT with the violated restriction active.

10 | SCALE Homogeneity, Euler, and multiplier shortcuts

10 Dot the KKT system with the allocation

For $Ax \leq b$ and $x \geq 0$, complementarity implies

$$x^\top \nabla f(x) = x^\top A^\top \lambda = \lambda^\top Ax = \lambda^\top b.$$

Hence the compact KKT system contains the useful identity

$$x^\top \nabla f(x) = \lambda^\top b. \quad (4)$$

If f is homogeneous of degree r , Euler's theorem turns (4) into

$$r f(x^*) = \lambda^{*\top} b.$$

With one binding budget $p \cdot x \leq m$,

$$\lambda^* = \frac{rV(m)}{m}.$$

This agrees with the envelope theorem because scaling income by t scales the value by t^r whenever the feasible set scales linearly.

Cobb–Douglas in two lines

Let $U(x) = \prod_i x_i^{a_i}$ and $r = \sum_i a_i$. Since logarithms preserve the optimizer, maximize $\sum_i a_i \log x_i$. Interior conditions give

$$\frac{a_i}{x_i} = \tilde{\lambda} p_i.$$

Multiplying by x_i and summing yields

$$r = \tilde{\lambda} m, \quad x_i^* = \frac{a_i}{r} \frac{m}{p_i}.$$

If the objective is replaced by $\phi(f)$ with $\phi' > 0$, the optimizer is unchanged but the multipliers are not. Under the same constraint normalization,

$$\tilde{\lambda} = \phi'(f(x^*))\lambda.$$

For $\phi(f) = \log f$, the original wealth multiplier is $\lambda = f(x^*)\tilde{\lambda}$. Map back before reporting an envelope effect.

For

$$\max_{c, x \geq 0} \{c + v(x) : c + p \cdot x \leq m\},$$

if $c^* > 0$, its stationarity condition pins down $\lambda = 1$. Hence

$$v_i(x^*) \leq p_i, \quad x_i^* [v_i(x^*) - p_i] = 0.$$

Used activities equate marginal benefit to price. If instead $c^* = 0$, the shortcut fails and $\lambda \geq 1$; the wealth boundary must remain explicit.

11 | DUALIZE

Many goods, one shadow price

11 The water-filling reduction

For a separable resource-allocation problem

$$\max_{x \geq 0} \sum_{i=1}^n u_i(x_i) \quad \text{subject to} \quad \sum_{i=1}^n p_i x_i \leq m,$$

fix a tentative shadow price λ . Each choice can then be solved independently:

$$x_i(\lambda) \in \arg \max_{x_i \geq 0} \{u_i(x_i) - \lambda p_i x_i\}.$$

For differentiable, strictly concave u_i ,

$$x_i(\lambda) = \begin{cases} 0, & u'_i(0+) \leq \lambda p_i, \\ (u'_i)^{-1}(\lambda p_i), & u'_i(0+) > \lambda p_i. \end{cases}$$

Now solve only the scalar resource-clearing equation

$$\sum_i p_i x_i(\lambda) = m.$$

The left side is nonincreasing in λ and strictly decreasing on the relevant interior segments, so bisection is reliable when a closed form is unavailable. Corners are selected automatically.

If

$$u_i(x_i) = a_i x_i - \frac{1}{2} b_i x_i^2, \quad b_i > 0, \quad 0 \leq x_i \leq \bar{x}_i,$$

then

$$x_i(\lambda) = \min \left\{ \bar{x}_i, \max \left\{ 0, \frac{a_i - \lambda p_i}{b_i} \right\} \right\}.$$

An activity enters when $\lambda < a_i/p_i$ and hits its cap when the unclipped formula exceeds $\bar{x}_i$. The regime map is encoded in these thresholds.

The dual problem is

$$\min_{\lambda \geq 0} \left\{ \lambda m + \sum_i \sup_{x_i \geq 0} [u_i(x_i) - \lambda p_i x_i] \right\}.$$

For every feasible allocation and every $\lambda \geq 0$, the expression in braces is an upper bound on attainable utility. Matching it with the value of a feasible allocation gives a zero-gap proof of global optimality. Separable primal choices become independent at a fixed price; the minimizing shadow price makes aggregate resource use consistent. KKT links the two views.

12 | SCREEN

Predict zero choices before solving

12 The entry test: would a missing activity pay for itself?

Assume KKT multipliers exist for the differentiable concave problem

$$\max_{x \geq 0} f(x) \quad \text{subject to} \quad Ax \leq b,$$

with Lagrangian $L = f(x) + \lambda^\top (b - Ax)$ and $\lambda \geq 0$. At shadow prices λ , define activity i 's **entry surplus**

$$E_i(x, \lambda) = \frac{\partial f(x)}{\partial x_i} - A_{\cdot i}^\top \lambda.$$

The first term is marginal benefit; the second is the shadow value of every resource required by one unit of the activity. KKT gives

$$E_i \leq 0, \quad x_i E_i = 0.$$

$$\begin{aligned} E_i > 0 &\implies \text{the proposed zero cannot be optimal,} \\ E_i < 0 &\implies x_i = 0 \text{ in any certified solution using these multipliers,} \\ E_i = 0 &\implies \text{first-order ambiguity: inspect curvature and degeneracy.} \end{aligned}$$

The exclusion margin $M_i = -E_i = A_{\cdot i}^\top \lambda - f_i(x)$ measures how far the activity is from entry. Its sign is invariant to positive rescaling of the objective, but its magnitude depends on units.

Restricted solve, full certificate

Choose a plausible active set S , impose $x_{S^c} = 0$, and solve the smaller problem. Then test each omitted activity $j \notin S$:

$$f_j(x^S) \leq A_{\cdot j}^\top \lambda.$$

If the included choices pass KKT and *one admissible* multiplier vector prices every omitted activity out, the restricted solution satisfies the full KKT system and is globally optimal. When restricted-problem multipliers are nonunique, one multiplier that fails is not enough to reject the allocation; look for a common multiplier vector that passes all entry tests.

For $u_i(x_i) = a_i x_i - \frac{1}{2} \kappa_i x_i^2$, with $\kappa_i, p_i > 0$ and one budget,

$$x_i(\lambda) = \frac{[a_i - \lambda p_i]_+}{\kappa_i}, \quad x_i = 0 \iff \frac{a_i}{p_i} \leq \lambda.$$

Sort the reservation values a_i/p_i . As resources tighten, λ rises and activities exit in threshold order; ties enter or exit together. If the budget is slack, $\lambda = 0$.

An infinite one-sided marginal at zero rules out a corner when a small entry is feasible and shadow cost is finite. A strictly dominated activity can be removed before solving. With complementarities or other interactions, however, the marginal value at the origin can mislead: solve the restricted incumbent bundle first and evaluate the entry surplus there.

13 | PROJECT

Quadratic KKT is geometry plus linear algebra

13 Complete the square before differentiating

For Q positive definite,

$$f(x) = c + a^\top x - \frac{1}{2}x^\top Qx = C - \frac{1}{2}(x - \bar{x})^\top Q(x - \bar{x}), \quad \bar{x} = Q^{-1}a.$$

Thus $\bar{x}$ is the unconstrained bliss point, and over a closed convex feasible set the constrained optimizer is its projection in the metric generated by Q . Active constraints are the faces hit by that projection.

For the reduced quantity-quality objective, put

$$a = 6 - p, \quad b = 4 - k.$$

Then

$$W(n, H) = y + \frac{1}{2}a^2 + \frac{1}{4}b^2 - \frac{1}{2} \left[(n - a)^2 + 2 \left(H - \frac{b}{2} \right)^2 \right].$$

The bliss point is therefore

$$(n_0, H_0) = \left(6 - p, \frac{4 - k}{2} \right).$$

The three demographic snapshots are different weighted projections of this point onto the subsistence set and quality cone.

Once a face is guessed, solve one linear system

Let the proposed active constraints be $Bx = d$. Stationarity and feasibility are

$$Qx + B^\top \lambda = a, \quad Bx = d,$$

or

$$\begin{pmatrix} Q & B^\top \\ B & 0 \end{pmatrix} \begin{pmatrix} x \\ \lambda \end{pmatrix} = \begin{pmatrix} a \\ d \end{pmatrix}. \quad (5)$$

If $BQ^{-1}B^\top$ is invertible, the Schur complement gives

$$\lambda = (BQ^{-1}B^\top)^{-1}(BQ^{-1}a - d).$$

Equation (5) produces a candidate, not an automatic answer. Check $\lambda \geq 0$, every inactive slack, all omitted-choice residuals, and possible dependence among active constraints.

14 | MAP Ratios, rays, and exact regime boundaries

14 Turn a cone into a one-dimensional problem

In the quantity-quality model, when $n > 0$ define

$$q = \frac{H}{n} \in [s, 4].$$

Using $a = 6 - p$ and $b = 4 - k$, the gain relative to the apex becomes

$$W(n, qn) - W(0, 0) = (a + bq)n - \left(\frac{1}{2} + q^2\right)n^2.$$

Conditional on the ray q , the unconstrained choice is

$$n_0(q) = \frac{[a + bq]_+}{1 + 2q^2}. \quad (6)$$

The subsistence constraint truncates this at $n_{\max}(q)$. For $q > 0$,

$$n_{\max}(q) = \frac{-(p + kq) + \sqrt{(p + kq)^2 + 2q^2(y - 1)}}{q^2},$$

while $n_{\max}(0) = (y - 1)/p$. Hence a two-choice nonlinear problem reduces to a bounded search over $q \in [s, 4]$, with

$$n^*(q) = \min\{n_0(q), n_{\max}(q)\}.$$

If small positive choices are resource-feasible, the apex is optimal exactly when

$$\max_{q \in [s, 4]} (a + bq) \leq 0.$$

Because the expression is linear in q , this becomes

$$a + bs \leq 0 \quad (b < 0), \quad a + 4b \leq 0 \quad (b \geq 0).$$

This is a global corner test obtained from feasible directions, not from solving for multipliers.

Solve a candidate active set symbolically. Convert each feasibility, multiplier, and reduced-cost check into a parameter inequality. Their intersection is the candidate's regime. A boundary occurs when one audit inequality becomes an equality; cross it by changing that one status and resolving. This avoids blind enumeration of all 2^{n+m} active sets.

EXAMPLE 1 Expected-utility portfolio choice**15 From a heuristic FOC to a corner-safe method**

A consumer has wealth $w = 24$. She invests $\alpha \geq 0$ dollars in a risky asset and $\beta \geq 0$ dollars in a safe asset:

$$\alpha + \beta \leq 24.$$

The safe gross return is 1. The risky gross return is $1/2$ in state L and $7/4$ in state H , each with probability $1/2$. Bernoulli utility is $u(c) = \sqrt{c}$. State consumption is

$$c_L = \frac{\alpha}{2} + \beta, \quad c_H = \frac{7\alpha}{4} + \beta,$$

so expected utility is

$$F(\alpha, \beta) = \frac{1}{2}\sqrt{c_L} + \frac{1}{2}\sqrt{c_H}.$$

Step 1: write the marginal utilities

$$F_\alpha = \frac{1}{8\sqrt{c_L}} + \frac{7}{16\sqrt{c_H}}, \quad F_\beta = \frac{1}{4\sqrt{c_L}} + \frac{1}{4\sqrt{c_H}}.$$

Step 2: write compact KKT

Let $\lambda \geq 0$ be the wealth multiplier. Then

$$\begin{aligned} 0 &\leq \alpha \perp \lambda - F_\alpha \geq 0, \\ 0 &\leq \beta \perp \lambda - F_\beta \geq 0, \\ 0 &\leq \lambda \perp 24 - \alpha - \beta \geq 0. \end{aligned}$$

The budget binds because adding one dollar to the safe asset raises consumption in both states. Hence $\alpha + \beta = 24$ and $\lambda > 0$ at the solution.

Step 3: make an active-set guess

Guess that both assets are held: $\alpha > 0$ and $\beta > 0$. Complementarity then gives

$$F_\alpha = F_\beta = \lambda, \quad \alpha + \beta = 24.$$

The guess will not be accepted until every sign is checked.

EXAMPLE 1 Solve, audit, and interpret**Step 4: solve only the implied equalities**

Subtract the two asset conditions:

$$F_\alpha - F_\beta = -\frac{1}{8\sqrt{c_L}} + \frac{3}{16\sqrt{c_H}} = 0.$$

Equivalently,

$$3\sqrt{c_L} = 2\sqrt{c_H}.$$

Since the budget binds, $\beta = 24 - \alpha$ and therefore

$$c_L = 24 - \frac{\alpha}{2}, \quad c_H = 24 + \frac{3\alpha}{4}.$$

Both square roots are nonnegative, so squaring is safe:

$$9\left(24 - \frac{\alpha}{2}\right) = 4\left(24 + \frac{3\alpha}{4}\right).$$

Thus $\alpha^* = 16$ and $\beta^* = 8$.

$$(\alpha^*, \beta^*) = (16, 8), \quad (c_L^*, c_H^*) = (16, 36),$$

$$F^* = \frac{1}{2}\sqrt{16} + \frac{1}{2}\sqrt{36} = 5, \quad \lambda^* = \frac{5}{48}.$$

Step 5: audit the candidate

At $(c_L, c_H) = (16, 36)$,

$$F_\alpha = \frac{1}{32} + \frac{7}{96} = \frac{5}{48}, \quad F_\beta = \frac{1}{16} + \frac{1}{24} = \frac{5}{48}.$$

Both asset holdings and λ are positive, the budget slack is zero, and both stationarity residuals are zero. All complementary products vanish. The payoff vectors are linearly independent, so strict concavity of square-root utility makes F strictly concave in (α, β) on the relevant domain. The verified KKT point is therefore the unique global maximizer.

Step 6: read the economics

Because a one-dollar shift from the safe asset to the risky asset changes the state payoff by $\tilde{R} - 1$,

$$F_\alpha - F_\beta = \mathbb{E}\left[u'(c)(\tilde{R} - 1)\right].$$

Thus the familiar expected-marginal-utility rule is exactly the interior KKT condition. The envelope theorem gives $V'(24) = \lambda^* = 5/48$: one more dollar of initial wealth raises maximal expected utility by approximately 0.1042.

EXAMPLE 1 Corners and a reusable generalization**16 Why KKT is safer than an interior heuristic**

Suppose instead that the consumer is risk neutral, $u(c) = c$. The risky asset's expected gross return is

$$\mathbb{E}[\tilde{R}] = \frac{1}{2} \left(\frac{1}{2} + \frac{7}{4} \right) = \frac{9}{8} > 1.$$

The optimum is the corner $(\alpha^*, \beta^*) = (24, 0)$. Take $\lambda^* = 9/8$. Then

$$\lambda^* - F_\alpha = 0, \quad \lambda^* - F_\beta = \frac{1}{8} > 0.$$

The second inequality is the correct condition at the zero safe holding. Forcing equality would incorrectly rule out the optimum.

At $\beta = 0$, moving one dollar from the risky asset to the safe asset lowers expected consumption by $1/8$. The unused asset has marginal value below the shadow value of wealth. KKT records this as an inequality.

The same square-root problem at general wealth

For any $w > 0$, the interior calculation gives

$$\alpha^*(w) = \frac{2w}{3}, \quad \beta^*(w) = \frac{w}{3},$$

$$c_L^*(w) = \frac{2w}{3}, \quad c_H^*(w) = \frac{3w}{2}.$$

The indirect value and multiplier are

$$V(w) = \frac{5\sqrt{w}}{2\sqrt{6}}, \quad V'(w) = \lambda^*(w) = \frac{5}{4\sqrt{6}\sqrt{w}}.$$

At $w = 24$, these formulas reproduce the solution on page 10.

For assets with payoff vectors r_i and holdings $x_i \geq 0$, an interior holding satisfies

$$\mathbb{E}[u'(c)r_i] = \lambda.$$

An excluded asset satisfies $\mathbb{E}[u'(c)r_i] \leq \lambda$. When one asset is the numeraire payoff, subtracting its equality gives the familiar excess-return condition. KKT adds the correct inequality at a corner.

EXAMPLE 2 A Galor-inspired quantity-quality model**17 A static laboratory for three regimes**

A household chooses parental consumption c , a divisible fertility index $n \geq 0$, and total human-capital investment H . If $n > 0$, define quality per child by $q = H/n$. Preferences are

$$U(c, n, H) = c + 6n - \frac{1}{2}n^2 + 4H - \frac{1}{2}H^2.$$

The constraints are

$$c + pn + kH + \frac{1}{2}H^2 \leq y, \quad c \geq 1, \quad sn \leq H \leq 4n.$$

Symbol	Meaning
y	household income
p	direct resource cost of fertility
k	linear price of total human-capital investment
s	minimum feasible quality per child
$c \geq 1$	subsistence consumption requirement
$sn \leq H \leq 4n$	quality cone: $s \leq q \leq 4$ whenever $n > 0$

The cone forces $H = 0$ when $n = 0$, so q is undefined at zero fertility. It also implies $H \geq 0$ whenever $s \geq 0$; no separate $H \geq 0$ multiplier is needed.

Because utility is increasing in c , the budget binds. Substitute

$$c = y - pn - kH - \frac{1}{2}H^2$$

to obtain the reduced objective

$$W(n, H) = y + (6 - p)n - \frac{1}{2}n^2 + (4 - k)H - H^2.$$

The reduced slacks are

$$R(n, H) := y - 1 - pn - kH - \frac{1}{2}H^2, \quad H - sn, \quad 4n - H.$$

The quantity-quality tradeoff and regime interpretation are inspired by unified growth theory. The quadratic preferences, convex education cost, quality cone, and exact zero-fertility corner are pedagogical extensions designed to isolate KKT logic; they are not claims about the exact specification in Galor's model.

EXAMPLE 2 Reduce once, then reuse the KKT system

Let $\eta, a, b \geq 0$ be the multipliers on $R, H - sn$, and $4n - H$. Treat n as nonnegative. Treat H as sign-free because its nonnegativity is already implied by the cone. Define

$$D_n = (6 - p) - n - \eta p - sa + 4b,$$

$$D_H = (4 - k) - 2H - \eta(k + H) + a - b.$$

$$0 \leq n \perp -D_n \geq 0, \quad D_H = 0,$$

$$0 \leq \eta \perp R(n, H) \geq 0,$$

$$0 \leq a \perp H - sn \geq 0,$$

$$0 \leq b \perp 4n - H \geq 0.$$

The reduced objective is strictly concave. The subsistence slack R is concave, so its superlevel set is convex; the cone is affine. Under Slater's condition, KKT is necessary, and any verified KKT point is sufficient. Strict concavity gives a unique primal allocation, though multipliers may be nonunique.

Mapping back to the original budget multiplier

Let λ be the multiplier on the original budget and μ the multiplier on $c - 1 \geq 0$. The original c stationarity condition is

$$1 - \lambda + \mu = 0.$$

After substitution, $\eta = \mu$, so

$$\boxed{\lambda = 1 + \eta.} \tag{7}$$

The envelope theorem gives $V_y = \lambda$. Thus η is the extra marginal value generated when the subsistence floor binds; it is not itself the original budget multiplier.

Substitution simplifies algebra, but it can hide multipliers. Always record how the reduced and original systems correspond. Here forgetting the “1+” in (7) would understate the marginal value of income.

EXAMPLE 2 Regimes I and II**Regime I: very-low-income, quantity-dominant**

Set $(y, p, k, s) = (5, 1, 5, 0)$. Guess $n > 0$, $R = 0$, $H = 0$, and the upper quality constraint slack. The active-set equalities are

$$\begin{aligned} D_n &= 5 - n - \eta = 0, \quad D_H = -1 - 5\eta + a = 0, \\ R &= 4 - n - 5H - \frac{1}{2}H^2 = 0, \quad b = 0. \end{aligned}$$

They give

$$\begin{aligned} (c^*, n^*, H^*, q^*) &= (1, 4, 0, 0), \\ (\eta^*, a^*, b^*) &= (1, 6, 0), \quad \lambda^* = 1 + \eta^* = 2. \end{aligned}$$

The budget, subsistence floor, and lower cone facet bind. The upper-cone slack is $4n - H = 16 > 0$, so $b = 0$ is correct. The positive fertility choice has $D_n = 0$, and $D_H = 0$. Every multiplier is nonnegative.

The household has only 4 units above subsistence. With high education cost $k = 5$ and no minimum-quality requirement, all of that margin goes to fertility. The income multiplier 2 exceeds the direct marginal utility of consumption because extra income also relaxes the binding subsistence restriction in the reduced problem.

Regime II: modern quantity-quality balance

Set $(y, p, k, s) = (20, 4, 2, 0)$. Guess $n, H > 0$ and all reduced constraints slack, so $\eta = a = b = 0$. Then

$$D_n = 2 - n = 0, \quad D_H = 2 - 2H = 0.$$

Hence

$$(c^*, n^*, H^*, q^*) = \left(\frac{19}{2}, 2, 1, \frac{1}{2} \right), \quad (\eta^*, a^*, b^*) = (0, 0, 0), \quad \lambda^* = 1.$$

The unused-income-above-subsistence slack is $R = 17/2$, and the two cone slacks are 1 and 7. The active-set guess passes every check.

EXAMPLE 2 Regime III and the exact-zero corner**Regime III: high quality cost with a minimum standard**

Set $(y, p, k, s) = (20, 4, 6, 2)$. At $n = H = 0$, the reduced subsistence slack is $R = 19$, so $\eta = 0$. Both cone facets bind at the apex. One valid multiplier choice is

$$b = 0, D_H = -2 + a = 0 \implies a = 2.$$

For fertility,

$$-D_n = -[2 - 2a] = 2 > 0.$$

This strict residual certifies the zero choice $n = 0$.

$$(c^*, n^*, H^*) = (20, 0, 0), q^* \text{ is undefined,}$$

$$(\eta^*, a^*, b^*) = (0, 2, 0), \lambda^* = 1.$$

The global comparison is especially transparent. Since $H \geq 2n$,

$$\begin{aligned} W(n, H) - W(0, 0) &= 2n - \frac{1}{2}n^2 - 2H - H^2 \\ &\leq -2n - \frac{9}{2}n^2 < 0 \text{ for every } n > 0. \end{aligned}$$

Thus no positive-fertility allocation can beat the apex.

The primal allocation is unique, but the cone multipliers are not. In fact,

$$b \in [0, 1], a = b + 2$$

works, together with a nonnegativity multiplier on n equal to $2 - 2b$ in the full system. At $b = 0$, the upper cone facet is binding but has zero multiplier. This is the promised example showing that binding does not imply a positive multiplier.

What changes across the three snapshots?

These regimes vary income and relative costs jointly; they are not a claim that income alone determines fertility. The exercise shows how KKT distinguishes an interior balance, a binding subsistence regime, and a genuine zero-choice corner.

MASTERY

From correct calculations to reliable judgment

18 The three-pass protocol

Pass	Questions to answer
1. Economize	What is monotone? What must bind? What can be ruled out by dominance, units, symmetry, or an Inada condition? Is there a better coordinate system, a useful ratio, a sharp bound, or a bliss point?
2. Solve	What are the primal–dual pairs? Which choices are positive? Which constraints bind? What equalities does that active set imply?
3. Certify	Does the candidate pass PDCS–GE below? Where is the active set valid in parameter space, and what do the multipliers mean?

P	primal feasibility	D	dual feasibility
C	complementary products	S	stationarity and zero-choice residuals
G	global theorem	E	economics, units, and envelope effects

The global certificate students should be able to reproduce

Let y be any feasible allocation. For concave differentiable f , KKT gives $\nabla f(x^*) = A^\top \lambda - r$, with $r \geq 0$, $x^* \perp r$, and $\lambda \perp b - Ax^*$. Therefore

$$\begin{aligned} f(y) - f(x^*) &\leq \nabla f(x^*)^\top (y - x^*) \\ &= \lambda^\top (Ay - b) - r^\top y \leq 0. \end{aligned}$$

This is why the certificate proves global optimality: feasibility makes both remaining terms nonpositive.

Rotate one drill at the start of class:

- translate prose into nonnegative slacks and name every unit;
- predict the active set before calculating;
- diagnose the first incorrect line in a polished but false solution;
- reverse KKT: infer parameter restrictions from a proposed certificate;
- solve one problem by substitution or geometry, then certify it by KKT;
- cross a parameter threshold and identify the single status that changes;
- obtain a comparative static from a multiplier without re-solving;
- defend a zero-choice inequality orally in sixty seconds.

Mastery means solving and auditing a two-choice, two-constraint concave problem in about fifteen minutes, deriving its parameter regimes, and explaining every multiplier without relying on memorized sign rules.

TEAR-OUT SHEET

Economic KKT at a glance

$$\max_{x \geq 0} f(x) \text{ s.t. } s(x) \geq 0, h(x) = 0.$$

$$D = \nabla f + J_s^\top \lambda + J_h^\top \nu.$$

$$0 \leq x \perp -D \geq 0,$$

$$0 \leq \lambda \perp s(x) \geq 0,$$

$$h(x) = 0.$$

$\lambda \geq 0$; equality multiplier ν is free.

For $Ax \leq b$,

$$0 \leq x \perp A^\top \lambda - \nabla f \geq 0,$$

$$0 \leq \lambda \perp b - Ax \geq 0.$$

Entry surplus: $E_i = \nabla_i f - (A^\top \lambda)_i$. Positive choice: $E_i = 0$; zero choice: $E_i \leq 0$; $E_i > 0$: enter. Positive multiplier: binding. Binding: multiplier may be zero.

1. Normalize each inequality as slack ≥ 0 .
2. Guess positive choices and binding constraints.
3. Solve the implied equalities.
4. Check original feasibility.
5. Check multiplier signs.
6. Check residuals at zero choices.
7. Check slack constraints have zero multipliers.
8. Check every complementary product.
9. Invoke the correct global theorem.
10. Interpret units and envelope effects.

1. Mixing $s \geq 0$ and $g \leq 0$ conventions.
2. Imposing equality at a zero choice.
3. Claiming binding implies a positive multiplier.
4. Dividing by a choice before proving it is positive.
5. Defining H/n when $n = 0$.
6. Squaring without checking signs or extraneous roots.
7. Claiming every KKT point is global without concavity.
8. Treating Slater as an existence theorem.
9. Inferring unique multipliers from strict concavity.
10. Using the envelope theorem before verifying the solution.

At a differentiability point,

$$V_\theta = f_\theta + \lambda^\top s_\theta + \nu^\top h_\theta.$$

For $Ax \leq b$, $V_{b_j} = \lambda_j$. For a consumer budget, $v_m = \lambda$ and $v_{p_i} = -\lambda x_i$.

Further reading

Boyd and Vandenberghe, *Convex Optimization*, chapters 5 and 10 ([official materials](#)). Ferris and Pang (1997), “Engineering and Economic Applications of Complementarity Problems” ([survey](#)). Rutherford (1995), “Extension of GAMS for Complementarity Problems Arising in Applied Economic Analysis” ([DOI](#)). Milgrom and Segal (2002), “Envelope Theorems for Arbitrary Choice Sets” ([paper](#)). Galor (2005), “From Stagnation to Growth: Unified Growth Theory” ([DOI](#)).

Prepared for ECON 803, Fall 2026. This booklet uses a single maximization/slack convention throughout. When consulting another source, translate its convention before comparing signs.